

Lumbar Flexion and Extension

Screening Test	Description
 Lumbar flexion/extension	Client position: • Standing • Knees straight • Feet hip-width Type of motion: Active Target motion: • Client forward bends (flexion) and attempts to touch the toes. Client arches backward (extension) as far as possible while keeping the neck in a neutral position. Verbal instructions: • “Bend forward as far as you can and try to touch your toes. Then arch your back as far as you can while keeping your head, neck, and knees straight, and your feet in place.” Assessment: • Normal mobility: ○ Flexion: Client is able to touch toes or floor without compensation ○ Extension: Client arches back enough to where their shoulders pass their hip joints without compensation • Restricted flexion: Client is unable to reach the target or ROM, or displays compensation such as accessory motion in other segments of the spine (i.e., cervical extension) ○ Overactive/shortened: Rectus abdominus, internal/external obliques

Hip Extension, Hip Adduction, and Knee Flexion (Modified Thomas Test)

Screening Test	Description
 Hip extension, adduction, and knee flexion (modified Thomas test)	Client position: - Lying supine at the end of a table, both knees bent over the edge of the table Type of motion: Passive Target motion: - Client holds the nontesting knee to chest, putting the hip in a maximally flexed position. Lumbar spine and pelvis are flat on the table. Test leg is relaxed on the table. The corrective exercise specialist looks for three movements of the test leg. Repeat on the other side: - o Hip extension: Assesses psoas length - o Hip abduction angle: Assesses <i>tensor fascia latae</i> (TFL) length - o Knee flexion: Assesses rectus femoris length Verbal instructions: - "Sit on the edge of the table and lie back with both knees bent off the table/let your legs hang. Grab your left leg at your knee and pull it toward your chest. Keep your right leg on the table completely relaxed. Repeat that same sequence with the other leg." Assessment: - Normal mobility: - o Hip extension: The test-side hip and thigh lay flat on the table in line with the torso without compensation - o Hip adduction: Femur rests in a straight line with torso (not abducted) with no compensation - o Knee flexion: Knee hangs naturally at 90 degrees off the table edge without compensation - Restricted hip extension: Test thigh lifts off the table - o Overactive/shortened: Psoas and rectus femoris - Restricted hip adduction: Test thigh is abducted and not in line with torso - o Overactive/shortened: TFL - Restricted knee flexion: Test knee is slightly extended and not at 90 degrees - o Overactive/shortened: Rectus femoris

Hip Abduction and External Rotation (Adductor Test)

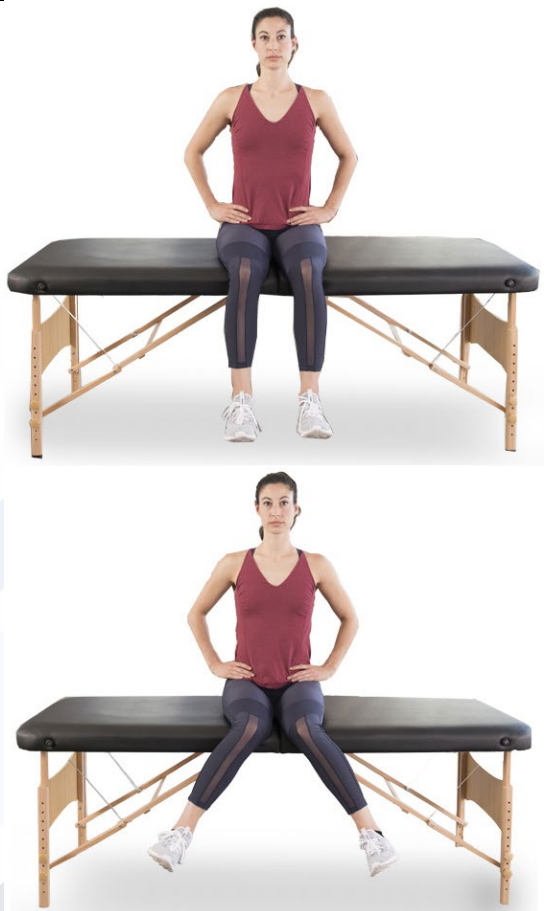
Screening Test	Description
 Hip abduction and external rotation (adductor test)	Client position: Lying supine on a table Type of motion: Passive Target motion: • Client lies supine with the test hip and knee bent to 45 degrees, with the foot placed on the medial portion of the opposite leg, between the thigh and shin as needed to achieve the correct angle • The nontesting leg is straight • Client passively relaxes the test leg to let it fall toward the table • Repeat on the opposite side Verbal instructions: • “Bend your right leg at the hip and knee to bring your foot against the opposite shin, and then allow your leg to passively relax. Repeat on your other side.” Assessment: • Normal mobility: Hip and knee on the testing side lay flat on the table without compensation • Restricted abduction: Client is unable to lay the test leg flat on the table, or displays impairments such as accessory motion in the opposite side of the pelvis (i.e., lifts off table) ○ Overactive/shortened: Hip adductor complex

Passive Hip Internal Rotation

Screening Test	Description
	Client position: Lying prone on a table with legs together Type of motion: Passive Target motion: • Client lies prone with both legs together so that knees are in line with the hips, and knees bent to 90 degrees of flexion • The client then passively allows the feet to spread apart (while maintaining bent knees), moving the hips into internal rotation Verbal instructions: • “Please bend your knees to 90 degrees. Then, relax and allow your ankles to fall away from each other and lower toward the table.” Assessment: • Normal mobility: Client can internally rotate the hip until the tibia (shins) are at a 45-degree angle to the table without compensation • Restricted internal rotation: Client is unable to reach the targeted mobility benchmark in either hip, or displays compensation such as knee extension, or lifting or abduction of the hips ○ Overactive/shortened: Piriformis, quadratus femoris, and gluteus maximus

Passive hip internal rotation

Seated Hip Internal and External Rotation

Screening Test	Description
	Client position: Seated at the edge of a table with both legs hanging over the side Type of motion: Active Target motion: • Client is seated with an upright posture, hands on hips, and both legs hanging over the side of the table. Client internally rotates one leg, as far as possible, without compensation through the LPHC. • Then, the client externally rotates the leg as far as possible (the other leg will need to internally rotate to accommodate the motion). This is repeated on the opposite side (internal rotation may be performed bilaterally). Verbal instructions: • “Sit at the edge of the table with an upright posture and hands on your hips. Internally rotate your right hip by moving your foot up and away from your opposite leg. Then, externally rotate your right hip by moving your foot up and toward your opposite leg. Try to keep your pelvis level and your back straight. Repeat this on your left side.” Assessment: • Normal mobility: Client is able to actively rotate the leg at the hip until the tibia (shin) is at an approximately 45-degree angle to the bench or table without compensation for both internal/external rotation • Restricted internal rotation: Client is unable to reach the targeted mobility benchmark in either hip, or displays impairments such as increased lateral flexion at the spine, hip hiking, or spinal flexion/extension ○ Overactive/shortened: Piriformis, gemellus superior/inferior, obturator internus/externus, quadratus femoris, and gluteus maximus • Restricted external rotation: Client is unable to reach the targeted mobility benchmark in either hip, or displays compensation such as increased lateral flexion at the spine, hip hiking, hip abduction, or spinal flexion/extension ○ Overactive/shortened: TFL, gluteus minimus and medius (anterior fibers), and hip adductors

Seated hip internal and external rotation